

A counterexample to the fivefold sumset inequality

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Abstract

We give finite sets of integers A, B satisfying $|A - B|^5 > |A + B|^5 |5B|$, where $5B$ is the fivefold sumset of B . The sets are base-193 encodings of nonnegative integer vectors in dimension 128, with coordinate sums bounded by 160 and 32, respectively. Stars and bars counts the sumsets, and a sum of 33 binomial terms counts the difference set. The strict comparison uses integer arithmetic and is independently reproduced by counting both positive and negative supports.

Keywords. sumset; difference set; integer simplex; exact certificate. **MSC 2020.** 11B75; 05A16.

1 The inequality and the construction

For finite nonempty subsets A, B of an abelian group, write $A \pm B = \{a \pm b : a \in A, b \in B\}$ and $hB = \{b_1 + \dots + b_h : b_1, \dots, b_h \in B\}$. Gyarmati, Hennecart and Ruzsa [1, equation (9), p. 177] asked for which integers h the inequality

$$|A - B| \leq |A + B| |hB|^{1/h} \quad (1)$$

holds for all such sets. Their discussion on p. 182 gives a counterexample for $h = 6$ and leaves $h = 4, 5$ unresolved. The result below gives an explicit counterexample to the $h = 5$ statement in that text; no priority or current-status claim is made.

For integers $m \geq 1$ and $r \geq 0$, define the integer simplex

$$V(m, r) = \left\{ x \in \mathbb{Z}_{\geq 0}^m : \sum_{i=0}^{m-1} x_i \leq r \right\}.$$

Its size is $\binom{m+r}{m}$. We use unequal radii in the classical lattice-simplex construction, used by Gyarmati, Hennecart and Ruzsa in their $h = 6$ example [1, p. 182].

Theorem 1. Let $\phi : \mathbb{Z}^{128} \rightarrow \mathbb{Z}$ be the additive map

$$\phi(x) = \sum_{i=0}^{127} 193^i x_i.$$

The finite integer sets

$$A = \phi(V(128, 160)), \quad B = \phi(V(128, 32))$$

satisfy

$$|A - B|^5 > |A + B|^5 |5B|. \quad (2)$$

*Developed with AI research agents. The proof is accompanied by two exact integer count formulas and a separate verifier.

2 Counting the vector sets

The count formulas hold for any pair of integer simplices.

Lemma 2. *For integers $m \geq 1$ and $a, b \geq 0$, let $X = V(m, a)$ and $Y = V(m, b)$. Then*

$$|X + Y| = \binom{m + a + b}{m}, \quad |5Y| = \binom{m + 5b}{m}, \quad (3)$$

$$|X - Y| = \sum_{j=0}^{\min(m,b)} \binom{m}{j} \binom{b}{j} \binom{m-j+a}{m-j}. \quad (4)$$

The difference count also equals

$$\sum_{j=0}^{\min(m,b)} \sum_{k=0}^{\min(m-j,a)} \binom{m}{j} \binom{m-j}{k} \binom{b}{j} \binom{a}{k}. \quad (5)$$

Proof. One has $V(m, a) + V(m, b) = V(m, a + b)$. To obtain the reverse inclusion, allocate up to a units of any vector's coordinate mass to the first summand; the remaining mass is at most b . Iterating this identity gives $5Y = V(m, 5b)$. Stars and bars proves (3).

For $z \in \mathbb{Z}^m$, put $(z_+)_i = \max(z_i, 0)$ and $(z_-)_i = \max(-z_i, 0)$. Then

$$z \in X - Y \iff \sum_i (z_+)_i \leq a \text{ and } \sum_i (z_-)_i \leq b.$$

Necessity follows from any representation $z = x - y$; sufficiency follows by taking $x = z_+$ and $y = z_-$. If z has exactly j negative coordinates, their positions can be chosen in $\binom{m}{j}$ ways. Their positive magnitudes have total at most b , giving $\binom{b}{j}$ choices. The other $m - j$ coordinates are nonnegative with total at most a , giving $\binom{m-j+a}{m-j}$ choices. This proves (4), including the case $j = 0$.

Alternatively, choose the j negative positions and then the k positive positions. Their nonzero magnitudes can be filled in $\binom{b}{j}$ and $\binom{a}{k}$ ways. Summing over the disjoint supports proves (5). $\square$

For the parameters of Theorem 1, set

$$D = \sum_{j=0}^{32} \binom{128}{j} \binom{32}{j} \binom{288-j}{128-j}, \quad (6)$$

$$S = \binom{320}{128}, \quad F = \binom{288}{128}. \quad (7)$$

These integers satisfy $D^5 > S^5 F$. The arithmetic can be checked with the following standalone Python code, using integer operations only:

```
from math import comb
D = sum(comb(128,j)*comb(32,j)*comb(288-j,128-j)
        for j in range(33))
S, F = comb(320,128), comb(288,128)
if D**5 <= S**5*F:
    raise ArithmeticError("fivefold comparison failed")
print("D^5 > S^5 F")
```

The accompanying verifier also evaluates (5) with factorials instead of the binomial-coefficient routine and checks the resulting integers against the stored values. Its small-set controls enumerate the sets themselves.

3 Encoding as integer sets

Proof of Theorem 1. Write $X = V(128, 160)$ and $Y = V(128, 32)$. Every coordinate of $X + Y$ lies in $[0, 192]$, and every coordinate of $5Y$ lies in $[0, 160]$. Base-193 uniqueness therefore makes ϕ injective on both sets.

Every coordinate of $X - Y$ lies in $[-32, 160]$. Adding 32 to each coordinate gives digits in $[0, 192]$ and adds the same integer $32 \sum_{i=0}^{127} 193^i$ to every image. Thus ϕ is injective on $X - Y$ as well. Additivity now gives

$$|A - B| = D, \quad |A + B| = S, \quad |5B| = F.$$

The integer comparison following (7) proves (2). □

References

- [1] K. Gyarmati, F. Hennecart and I. Z. Ruzsa, *Sums and differences of finite sets*, *Functiones et Approximatio Commentarii Mathematici* **37**(1) (2007), 175–186. <https://doi.org/10.7169/facm/1229618749>.